

Steps to upgrade AWS Oracle RDS

Upgrades are majorly of 2 types :

- **Major Version Upgrade**

Major Version Upgrade aren't applied automatically by AWS and this need a manual intervention, since the changes introduced may not be compatible with your existing Applications.

- **Minor Version Upgrade**

If you enable auto minor version upgrades on your DB instance, minor version upgrades occur automatically.

You need to ensure that the Application Team is aware of the Scheduled Maintenance Window of your Minor Version Upgrade, since AWS sometimes doesn't notify priorly for the DB Minor Version Upgrade.

AWS Link :https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_UpgradeDBInstance.Oracle.html

Step 1: We need to check the Option Groups and Parameter Groups before proceeding with the upgrade.

If your database is using Default Option and Parameter groups, you need not worry and proceed with your upgrade. If they are using Custom Option /Parameter groups, you need to create the same compatible with the new version and attach them to the database after upgrade by modifying your instance.

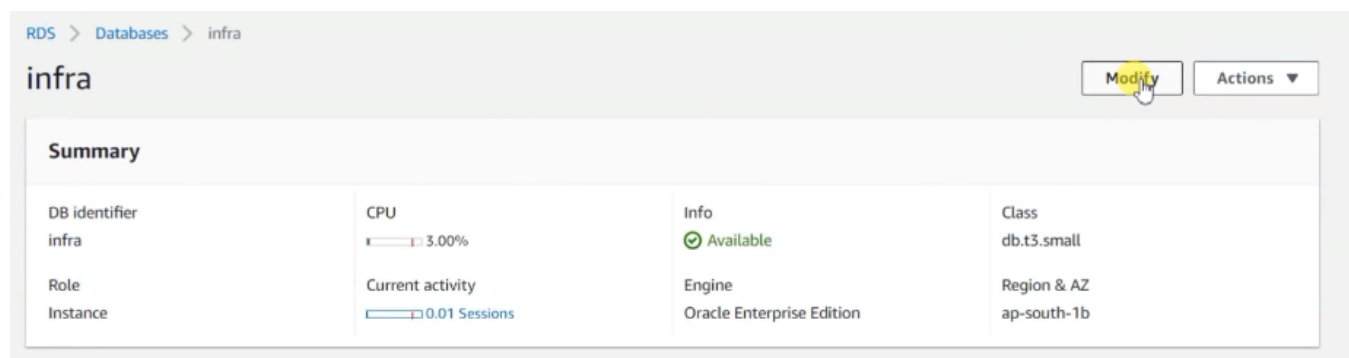
Step 2: If automatic backups aren't enabled, please ensure to take a snapshot of your database before proceeding with the change. If automatic backups are enabled, AWS will automatically take a backup of your database before upgrading

You can verify whether your databases automatic backups are enabled or not, by checking on the Backup-retention period. If the Backup Retention Period is greater than 0, then Automatic Backups are enabled for your database.

Step 3: Login to your database and execute Stats Gather

```
EXEC DBMS_STATS.GATHER_DICTIONARY_STATS;
```

Step 4: Now to upgrade your database engine, click on your database --> Modify



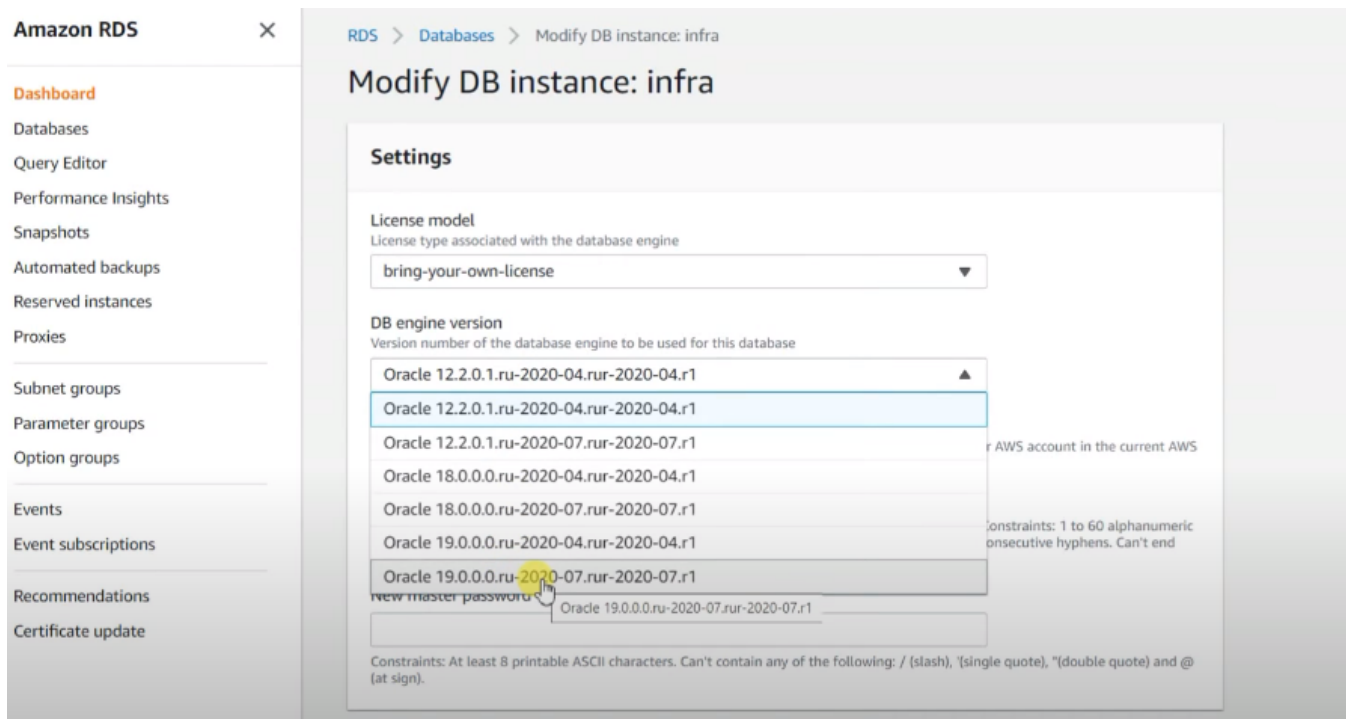
RDS > Databases > infra

infra Modify Actions ▼

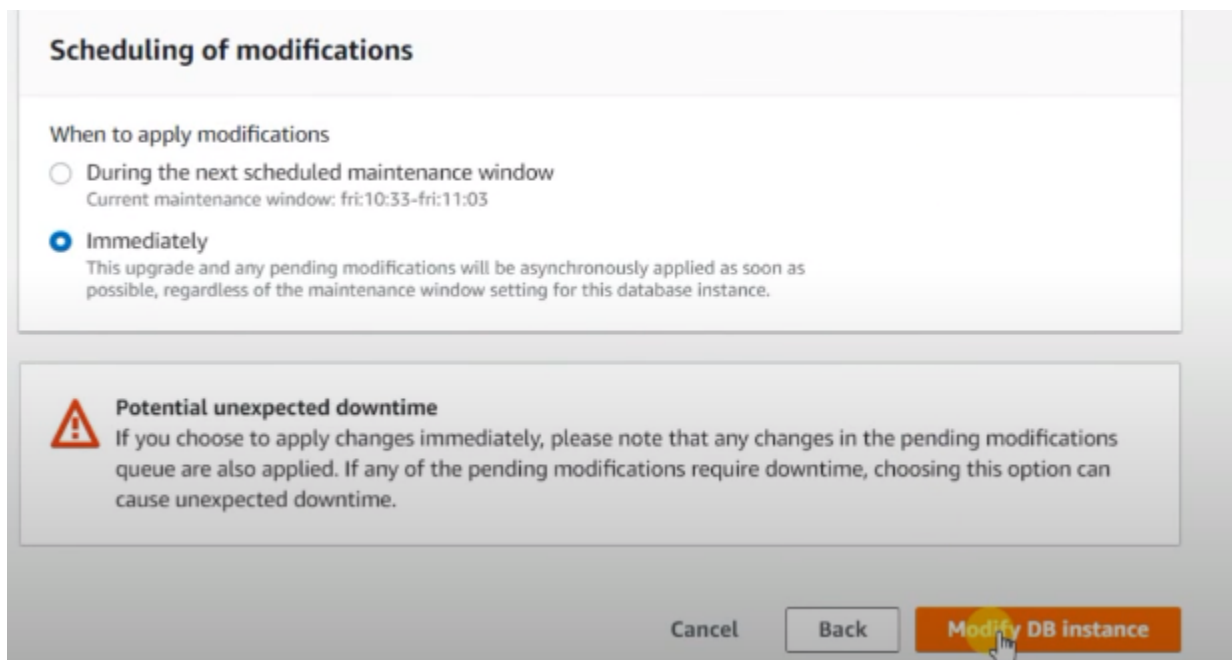
Summary

DB identifier infra	CPU 3.00%	Info Available	Class db.t3.small
Role Instance	Current activity 0.01 Sessions	Engine Oracle Enterprise Edition	Region & AZ ap-south-1b

Step 5: Change the **DB Engine Version** to 19c



Step 6: Click on **Apply Immediately** if you are planning to upgrade the database instantly, or else Click the other option to upgrade during the next scheduled Maintenance Window and **Modify DB Instance**



Step 7: Once you click on **Modify DB Instance**, a snapshot of your Database instance will be taken (if Backup retention period is more than 0 days), and then the Database will be upgraded.

NOTE : Once the database is upgraded, it won't be possible to revert back the changes. If you need the Older version of your database, you need to manually restore it from the snapshot. The DB Snapshot is taken before and after the upgrade activity.

Step 8: 8. Once the DB is upgraded, you can change the default parameter/option groups with the custom parameter/option groups compatible with the newer version by modifying the instance.